

**SinhalaJournal-LLM: A Style-Controlled Large Language
Model for Diverse Sri Lankan Newspaper Writing**

R26-SE-037

Project Proposal Report

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March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Advanced natural language processing techniques continue to underperform Sinhala news writing, especially when it comes to assignments that call for controlled rewriting for various journalistic styles. Existing solutions do not provide style-aware rewriting of Sinhala news content for newsroom use, even while they offer translation, typing help, or limited summarization. This component focuses on the Sinhala Rewriter within the broader SinhalaJournal-LLM project. This proposed component aims to rewrite a given Sinhala news passage into a target style like formal , editorial, sports, or youth oriented without changing the actual meaning and readability. The lack of a style-controlled Sinhala news rewriting system that is trained on locally obtained newspaper data and assessed for both style adherence and meaning preservation is the research gap. Corpus cleaning, Unicode normalization, paragraph segmentation, style schema design, hybrid labelling, gold rewrite pair generation, baseline prompt experiments, and parameter-efficient fine-tuning utilizing LoRA on a large language model capable of Sinhala are all included in the technique. A combination of automated and human-centered metrics, such as semantic similarity, style accuracy, fluency, and factual consistency, will be used for validation. A LoRA-adapted rewriting model, a curated style-transfer dataset for Sinhala journalism, and an interface-ready component that can be incorporated into the team's final editorial assistant are among the anticipated results. The component has commercial value as a potential SaaS/editorial productivity tool for newsrooms, digital publishers, and journalism education, as well as social value in enhancing consistency and accessibility in Sinhala news writing.

Keywords: journalism, low-resource language, LoRA fine-tuning, Sinhala NLP, text style transfer The AI

ACKNOWLEDGEMENT

I would like to convey my sincere appreciation to my research supervisor Prof. Nuwan Kodagoda and co-supervisor Ms. Poojani Gunathilake for their continuous guidance and support throughout this study. Their encouragement, constructive feedback, and insightful discussions greatly contributed to the development of this research.

I am also thankful to my project team members for their collaboration and shared efforts in planning and structuring the SinhalaJournal-LLM platform. Finally, I extend my appreciation to everyone who supported me with their valuable time and assistance during the completion of this proposal.

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LIST OF ABBREVIATIONS

API – Application Programming Interface

DSR – Design science research

GPU – Graphics Processing unit

LLM – Large language model

LoRA – Low -Rank Adaption

NLP – Natural Language Processing

PEFT – Parameter – Efficient Fine-Tuning

REST – Representational State Transfer

SaaS – Software as a Service

TF-IDF - Term Frequency – Inverse Document Frequency

UI – User Interface

WBS – Work breakdown Structure

Rouge - Recall-Oriented Understudy for Gisting Evaluation

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1. INTRODUCTION

1.1 Background and Context

For the purpose of rewriting, editing, and adapting the same news information to various publishing styles, Sinhala journalism still primarily relies on manual editorial work. In high-volume digital news contexts, where publishing quality is directly impacted by speed, consistency, and readability, this becomes especially difficult. Journalists, editors, digital publishers, and journalism students who depend on effective editorial tools to generate high quality news material are directly impacted by the issue. More than 17 million people in Sri Lanka speak Sinhala, however compared to high-resource language like English, there are still few sophisticated NLP tools available for Sinhala. Current solutions, such as keyboard programmes and translation systems, primarily facilitate typing or translation but lack the ability to rewrite in a news-room oriented style. Although there are research prototypes for tasks like grammar analysis and summarization, these systems are still disjointed and do not provide controlled rewriting across various journalistic styles. This component especially focuses on style rewriting within the larger SinhalaJournal-LLM system, allowing for the controlled conversion of Sinhala news material into various publishing styles while maintaining factual meaning.

1.2 Literature Review

In several high-resource languages, recent developments in Natural Language Processing (NLP) and Large Language Models (LLMs) have greatly enhanced automated text production, summarization, and language comprehension. However, low-resource languages like Sinhala do not have equal access to these technical advancements. Large-scale datasets, specialized language models, and domain-specific NLP tools that facilitate difficult tasks like newspaper writing are still lacking in Sinhala. Because of this, current multilingual language models frequently have trouble with contextual language usage, morphological complexity, and Sinhala grammar. Tasks like news story creation, grammar correction, summary, and style editing make this constraint very apparent.

By creating linguistic resources and specific models, several studies have tried to enhance Sinhala NLP capabilities. (2024) unveiled NSina, a sizable Sinhala news corpus with more than 500,000 items gathered from well-known Sinhala news websites. Benchmark activities including media identification, news classification, and headline creation are also included in the dataset. This dataset offers a crucial starting point for creating NLP models tailored to Sinhala news applications. In a similar vein, (2022) carried out a thorough examination of pretrained language models for the classification of Sinhala texts. Their work, called BERTifying Sinhala, showed that multilingual models like XLM-R, LaBSE, and LASER are greatly outperformed by monolingual Sinhala models based on RoBERTa. These results emphasize how crucial it is to modify language models to account for Sinhala's linguistic features.

According to the Survey of Publicly Available Sinhala Natural Language Processing Tools and Research (2024), which examined the larger field of Sinhala NLP research, Sinhala is still a language with few resources because there aren't many coordinated research projects, datasets, or tools available. The study highlights how numerous Sinhala NLP projects

function independently, leading to dispersed advancements in fields like machine translation, morphological analysis, and part-of-speech tagging. In order to expand the multilingual LLaMA architecture with Sinhala-specific tokenization and extra pretraining on a Sinhala corpus with more than 10 million sentences, the SinLlama model was presented more recently. This study showed that language-specific adaptation greatly enhances performance on Sinhala tasks and offers a possible foundational model for Sinhala generative systems in the future.

In addition to Sinhala-specific research, a number of studies have investigated text style transfer techniques, which are closely linked to the suggested system's style rewriting component. Style transfer is the process of changing a text's stylistic elements while maintaining its original semantic meaning. According to a recent assessment on text style transfer (2024), contemporary methods depend more and more on huge language models that can use quick engineering to accomplish zero-shot and few-shot style transformation. Large style-labeled datasets are not necessary for these models to rewrite text into various stylistic forms. However, the report also highlights the additional difficulties that low-resource languages present because of the scarcity of evaluation benchmarks and training data.

Effective methods for tailoring big language models to domain-specific tasks have also been investigated in recent studies. (2022) presented Low-Rank Adaptation (LoRA), a parameter-efficient fine-tuning technique that trains a limited number of low-rank matrices inside the transformer layers while freezing the majority of the pretrained model weights. This method allows for efficient domain adaptation while drastically lowering the number of trainable parameters and processing demands. LoRA is especially well-suited for low-resource settings where large-scale model training is impractical, according to later research on parameter-efficient fine-tuning methods.

Even while these studies offer insightful information, there are still a number of issues with the state of the field. First, rather than sophisticated generative tasks, the majority of Sinhala NLP research concentrates on fundamental language processing tasks like categorization, morphological analysis, or extractive summary. Second, while many current tools, like Google Translate and Sinhala keyboard apps, handle basic languages, they don't have domain-specific features for journalism writing. Third, due to a lack of training data, current multilingual LLMs frequently generate awkward or grammatically inconsistent Sinhala text. Lastly, it is challenging to reformat the same content into multiple journalistic styles, such as editorial, sports, or youth-oriented writing, while maintaining the original meaning because current systems hardly offer tools for explicit stylistic control.

Therefore, creating style-controlled language generation systems for Sinhala journalism still clearly needs investigation. By utilizing Sinhala news datasets, modifying a pretrained language model through LoRA-based fine-tuning, and permitting controlled rewriting across various journalistic styles, the suggested SinhalaJournal-LLM system seeks to overcome this constraint. This method supports more efficient and context-aware Sinhala news writing by combining recent developments in NLP, style transfer, and parameter-efficient model adaption.

1.3 Research Gap

Even though Sinhala Natural Language Processing (NLP) has improved significantly in recent years, there are still a number of substantial obstacles to the creation of sophisticated generative systems for Sinhala newspaper writing. Instead of focusing on sophisticated generative applications that facilitate actual editorial workflows, current research mostly concentrates on fundamental tasks like text classification, machine translation, or extractive summarization. For instance, a sizable Sinhala news corpus and a number of benchmark tasks for Sinhala NLP research, such as news classification and headline generation, were introduced via the NSina dataset.

In a similar vein, research like BERTifying Sinhala shows how well monolingual pretrained language models work for Sinhala text categorization tasks. However, rather than generating tasks that call for rewriting or transforming pre-existing information, these models are primarily intended for comprehension activities. Even more contemporary Sinhala language models, like SinLlama, are primarily concerned with enhancing downstream NLP performance and generic language modelling capabilities. Though they don't really handle domain-oriented applications like carefully rewriting news information into various journalistic styles, these models offer a basis for Sinhala text production.

Promising methods for changing textual style while maintaining semantic meaning have also been developed by research on text style transfer. However, the majority of style transfer techniques currently in use are mainly designed for high-resource languages like English and are predicated on the availability of big style-labelled datasets. Training efficient style transfer systems is difficult in low-resource languages such as Sinhala because there are very few style-labelled corpora.

Apart from research constraints, the publicly accessible language tools now in use only offer rudimentary Sinhala language processing capabilities. Users can get assistance with translation, typing, and spelling recommendations via programs like Google Translate and Sinhala keyboard utilities like Helakuru. However, rather than specialized writing duties, these systems are primarily intended for broad language support. Advanced features like context-aware grammatical correction, automatic headline production, and style-controlled news content rewriting are not offered by them. Because of this, when creating Sinhala news pieces, editors and journalists still mostly rely on manual editing procedures.

Additionally, the majority of NLP systems now in use for processing Sinhala news rely on either general-purpose multilingual language models or conventional extractive techniques. For instance, algorithms like TF-IDF and TextRank were employed in earlier studies on Sinhala news summarization to extract important sentences from articles. These methods are capable of identifying crucial information, but they are unable to produce content that is both stylistically appropriate and cohesive. Due to insufficient training on Sinhala-specific data, multilingual big language models also frequently generate inconsistent or unnatural Sinhala outputs.

By establishing a style-controlled rewriting system that can convert Sinhala news stories into several journalistic styles, including formal news reporting, editorial analysis, sports reporting, and youth-oriented writing, the proposed Sinhala Style Rewriter component seeks to close this gap. The system will enable controlled production of stylistically acceptable news content while preserving semantic consistency by utilizing Sinhala news corpora and

modifying pretrained language models using methods like Low-Rank Adaptation (LoRA). This method adds to the creation of useful AI tools for Sinhala news media and presents a novel application of style transfer techniques to Sinhala journalism writing.

2. OBJECTIVE

2.1 Main Objective

Developing a style-controlled Sinhala news text rewriting system that can convert news articles into various journalistic writing styles while maintaining the original semantic meaning and factual correctness is the main goal of this research component.

By allowing journalists and editors to automatically rewrite Sinhala news information into several styles, including formal news reporting, editorial commentary, sports reporting, and youth-oriented writing, the technology seeks to support newsroom operations. The component aims to enhance the effectiveness, consistency, and flexibility of Sinhala news content generation within the SinhalaJournal-LLM system by utilizing pretrained language models and parameter-efficient fine-tuning procedures.

2.2 Specific Objectives

Gathering and Preparing Datasets for Style Rewriting

- Gather news items in Sinhala from several trustworthy online news sources and create a structured dataset that can be used to train and assess the style-controlled rewriting system.
- By eliminating noise, normalizing Sinhala text, and classifying articles into pertinent groups, the gathered dataset is preprocessed and cleaned.
- Make sure the content is appropriate for various journalistic writing styles in order to prepare the dataset to allow style-based rewriting activities.

Recognizing and Examining Journalistic Writing Styles in Sinhala

- Determine the many Sinhala journalism writing styles, including youth-oriented language, formal news writing, editorial writing, and sports reporting.
- Examine the language usage, linguistic patterns, and stylistic traits connected to each writing style.
- Establish style standards that the rewriting system can employ to produce content that is appropriate in terms of style.

Optimizing a Pretrained Language Model for Rewriting Sinhala News

- Choose a pretrained language model that is suitable for tasks involving the creation of Sinhala text.
- To modify the model for style-controlled Sinhala news rewriting, use parameter-efficient fine-tuning techniques like Low-Rank Adaptation (LoRA).

- To enable efficient style transformation, train and tune the model using the provided Sinhala news dataset.

Assessment of the Style Rewriting Method

- Analyze the rewriting system's performance using both quantitative and qualitative indicators, such as linguistic fluency, style accuracy, and semantic similarity.
- Test the model's capacity to maintain meaning while modifying the text to fit the chosen journalistic style.
- Examine the system's performance in actual news rewriting situations.

3. METHODOLOGY

3.1 Research approach

This Research includes Design Science Research (DSR) approach combined with experimental evaluation. Design science Research focuses on designing and evaluating real world problem solving technological artifacts. In this project the artifact is SinhalaJournal-LLM and AI based language platform that designed to support to Sinhala article writing with headline and image generation , grammar correction , news summarization, and style rewriting. The target of this component is the Sinhala Style Rewriter, which enables rewriting of Sinhala articles into various journalistic styles while keeping the original meaning. The process includes several stages : Problem identification , data collection, data preprocessing, model development, system implementation. After gather Sinhala news articles , the data will be cleaned and structured to build a training dataset. A pretrained LLM will next be adapted through parameter efficient fine tuning to make Style controlled rewriting. At last the model will be evaluated using quantitative matrices and qualitative human evaluation.

3.2 Technical methods and algorithms

Large Language Models (LLMs) based on transformers and parameter-efficient fine-tuning methods constitute the foundation of the Sinhala Style Rewriter.

First all collected Sinhala news articles will go to preprocessing steps like Unicode normalization, removal of HTML tags, duplicate removal, and sentence segmentation. After that the dataset is clean and suitable for training. Next define the various journalistic writing styles like formal news writing, editorial style, sport reporting style, and youth-oriented style. Each segment will be labeled based on characteristics using manual annotation and source-based heuristics. The rewrite model follow the instruction-based generation approach , where the model get the original text together with the desired style. As the example the system prompt the model to rewrite a paragraph in formal style while keeping the original meaning. The efficiently adapt the model, this use LoRa (Low-Rank Adaptation).

LoRA. LoRA freezes most pretrained model weights and trains only small number of additional parameters enter to the transformer layer.

3.3 Tools and platforms with justification

Python has a large library for machine learning and natural language processing, it will be the main programming language. Python offers quicker development for NLP applications and simpler integration with AI frameworks when compared to languages like Java or C++.

The model will be trained using PyTorch, a deep learning platform. PyTorch provides robust interoperability with contemporary NLP libraries and flexible model construction. PyTorch offers dynamic computation graphs and easier debugging than TensorFlow, which are helpful

for experimentation. Additionally, pretrained language models will be accessed and LoRA-based fine-tuning will be implemented using the HuggingFace Transformers library. HuggingFace makes it much easier to modify huge language models for certain tasks by offering ready-to-use transformer structures and pretrained checkpoints. HuggingFace ensures interoperability with contemporary NLP research implementations while reducing development complexity and training time when compared to creating transformer models from scratch using raw deep learning frameworks.

Table 1 Tools and Platforms with Justification

Tool / Platform	Purpose	Justification
SinLlama	Base Large Language Model	Sinhala Adapted LLaMA based model trained on Sinhala corpora. It gives better language knowledge and generation for Sinhala compared to general multilingual models.
PEFT (LoRA)	Petameter efficient model adaption	Enables fine tuning or LLM s by training just a little number of parameters while holding the original model weights. Reduce the training cost.
PyTorch	Model training	Mostly used in learning frameworks that provides flexible model design , dynamic computation graphs and good NLP support.
Google Colab / GPU Server	Training model environment	Gives GPU resources for training and fine tuning models without needing expensive hardware.
HuggingFace Transformers	Model loading, tokenization, and fine-tuning	enables effective experimentation and integration of large language models by offering ready-to-use transformer architectures and pretrained checkpoints.
Rouge and Blue metrices	Model evaluation	NLP evaluation matrices used to measure text generation quality comparing with generated article output with texts.

3.4 Data collection and ethical considerations

The Sinhala news items gathered from publicly accessible online news sources, such as Divaina, Hiru News, Ada Derana, and Lankadeepa, will make up the dataset used in this study. Large amounts of Sinhala news covering topics including politics, business, sports, and worldwide news are published by these sources.

A headline, article body, category, timestamp, and URL are usually included in each article. These components offer enough data to create a dataset that can be used to train the style rewriting model. It is anticipated that the dataset will include between 500,000 and 1,000,000 Sinhala news items, which is sufficient to train NLP models specialized to a given topic. Automated web scraping programs will be used to collect data, and during preparation,

duplicate or corrupted records will be eliminated. The procedure of gathering data will also adhere to ethical standards. The dataset will only contain publicly accessible news stories from online news sources; it will not contain any sensitive or private user information. The information gathered will only be utilized for scholarly research, and original sources will be cited as needed.

3.5 Validation strategy and performance metrics

Both automated evaluation metrics and human evaluation will be used to assess the Sinhala Style Rewriter's performance.

First, whether the revised text retains the original text's meaning will be assessed using semantic similarity metrics. Successful content preservation is indicated by high similarity scores. Style accuracy will assess whether the chosen writing style is reflected in the output text. This may be assessed using style classification models or manual review. To make sure the generated text is grammatically correct and natural for Sinhala readers, readability and fluency will be assessed.

Lastly, a human review of the updated outputs' utility and readability will be carried out with journalism students or Sinhala speakers. These assessments will assist in determining whether the technology is capable of supporting actual newsroom writing responsibilities.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system, SinhalaJournal-LLM is an based language assistant platform that designed to support to write Sinhala articles. This system integrates NLP components that assist journalists and editors to improve their articles efficiently. The components are Headline and image generation, grammar correction, news summarization, and style rewriting. Each component works a specific task while sharing a common pretrained language model and dataset.

The mainly focus of this research is the Sinhala style Rewriter component, which allow rewriting Sinhala news content into different journalistic styles while keeping the original meaning. This component supports writing styles like formal, editorial, sports, children and youth writing. The architecture allow these components to work together within a shared preprocessing pipeline, enabling efficient text generation.

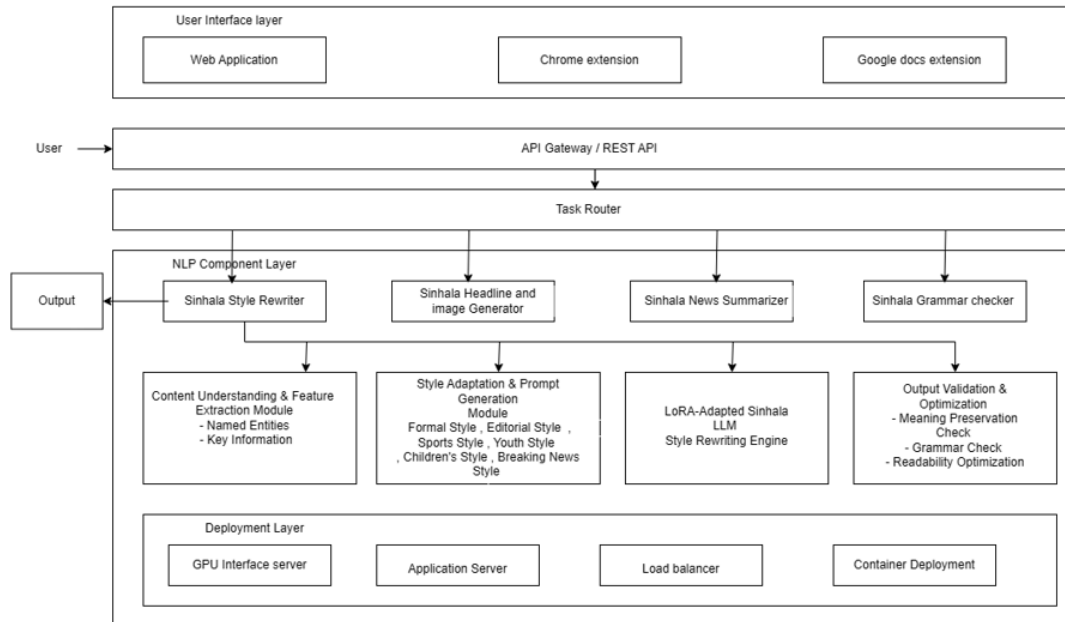


Figure 1 High Level System Architecture

The Sinhala Style Rewriter is the individual component implemented in this research. The main purpose is to transform Sinhala news articles into different writing styles while keeping the actual meaning. In the Style rewriter component consists of several sub components:

- **Input Text Processor** – Receive Sinhala News articles and prepare it for preprocessing.
- **Style selection module** – Allows users to select the writing style (formal, editorial,).
- **The LoRA-Adapted Language Model** is a pretrained transformer-based language model that adapted using LoRa for style-controlled text generation.
- **Output Refinement Module** – ensures grammatical correctness and readability before returning the generated text.

The Sinhala Style Rewriter connect with several components within the SinhalaJournal-LLM System. User interface gather the news article and the writing style from the user. This input is passed to the Preprocessing module, which performs text cleaning , Unicode normalization, and segmentation. After preprocessing , the cleaned text is going to the Core language Model Layer , Which give the pretrained language model. The Styke rewrite component uses this model with LoRa-based to give the rewritten output.

The system starts when a user submits a Sinhala news article using the interface. The article text is first processed by the preprocessing layer. In that removes unwanted characters, Normalize Sinhala Unicode text, and segments the article into suitable inputs. Next, the processed article text and user selected writing style are move to the Style rewrite module. The module uses the pretrained language model with style control details to generate rewritten article text. After generation, the output moves through the post processing stage to check readability and formatting consistency.

The architecture is designed to be scalable through its modular design. All the Natural Language Processing components share the same base language model and preprocessing pipeline. The use of LoRa Fine tuning also enhanced scalability by enabling good model adaption with lower computational requirements.

From the security , the system uses only Sinhala news data which are publicly available and does not process sensitive personal information. Input validation parts will be implemented to reduce malicious or invalid inputs. Access to trained models and data sets will be controlled to maintain system integrity and reliability.

5. USER REQUIREMENTS

5.1 Stakeholder identification

Several user groups engaged in Sinhala news writing and content creation are intended to be supported by the proposed SinhalaJournal-LLM system. Among the important stakeholders are:

- Journalists: This system mainly used by Journalists. They may swiftly create other versions of news items, Improve readability, and rewrite articles in different writing styles using the system.
- News Editors: By utilizing several style types, like formal or editorial, editors can use the system to evaluate and improve news material.
- Digital Media Organizations: The can be used by media organizations and online news platforms to improve productivity and keep a uniform writing style throughout published pieces.
- Journalism Students: By use this system as learning tool , journalism students can improve their writing skills and get a good understand of various news writing styles.
- Researchers and Developers: The system and dataset may be used by researchers studying AI and natural language processing to future testing and improve Sinhala NLP technology.

5.2 Functional requirements

The primary features that the Sinhala Style Rewriter component and the SinhalaJournal-LLM system must offer are outlined in the following functional criteria.

FR1: The system allow to users to input Sinhala news articles or text using the user interface.

FR2: The system preprocesses the incoming text by performing questions like cleaning, Unicode, normalization, and sentence segmentation.

FR3: The system allow to user to select a target writing style like formal , editorial , sports, youth , children or breaking news style .

FR4: The system will rewrite the input Sinhala article into the selected writing style while keeping the original mean.

FR5: The system produces grammatically correct Sinhala text outputs.

FR6: The system shall display the generated article into the user through the interface.

FR7: The system will allow to users to request and generate different rewritten versions of the same article with different styles.

FR8: Sinhala news datasets used to train the language model will be stored and processed by the system.

5.3 Non-functional requirements

The system's operational features and quality qualities are described by non-functional requirements.

Safety -The system shall ensure that user input articles are safe and protected from malicious and invalid data and access to the trained language model and dataset should be safe from unauthorized users.

Performance -The system should generate outputs within a few seconds for standard news article.

Usability – The system interface need to simple and user friendly with allowing journalists and editors to input articles and texts without high technical knowledge.

Reliability- The system should continuously generate meaningful Sinhala text outputs and reduce errors like grammatical mistakes or loss of important information.

Scalability – The architecture should allow new writing styles to be added in the future without need to redesign the whole system.

Sustainability - In order to allow for the independent updating and improvement of each component, including the Style Rewriter, the system should be built in a modular fashion.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The SinhalaJournal-LLM platform targets Sri Lankan digital media organizations, publishing houses, and online news portals that produce high volumes of Sinhala-language content. These organizations require automated support for grammar validation, headline generation, summarization, and style adaptation to improve editorial efficiency and consistency.

Secondary customer segments include:

- Universities and journalism institutes
- Government communication departments
- Digital marketing agencies
- Independent content creators and bloggers

The primary customer persona is a newsroom editor managing multiple daily publications who requires AI-assisted tools to accelerate content production while maintaining linguistic quality.

6.2 Value Proposition

SinhalaJournal-LLM provides an end-to-end AI-assisted newsroom platform that supports:

- Automated grammar correction
- Style-adaptive headline generation
- Controlled tone rewriting
- News summarization

Unlike standalone tools, the integrated architecture ensures that all generated outputs pass through a centralized grammatical validation layer, maintaining linguistic consistency across the platform.

Key measurable benefits include:

- 30–50% reduction in manual proofreading time
- Faster content production cycles
- Improved consistency in journalistic tone
- Reduced editorial workload

The platform addresses a critical technological gap, as no comprehensive AI-powered Sinhala newsroom assistant currently exists.

6.3 Revenue Model

The system follows a Software-as-a-Service (SaaS) model with recurring subscription revenue.

1. Institutional Subscription Plans

Media organizations can subscribe under tiered plans:

- Basic Plan – Limited monthly corrections and summarizations
- Professional Plan – Higher request limits with priority processing
- Enterprise Plan – Unlimited usage, API access, analytics, and dedicated support

Annual subscription packages may be offered at discounted rates.

2. API-Based Pricing

Third-party news platforms, content management systems, and mobile applications can integrate SinhalaJournal-LLM via APIs. Pricing may follow:

- Per-request billing
- Per-token usage pricing
- Custom enterprise API agreements

3. Educational Licensing

Universities and institutes may receive discounted academic licenses to support journalism education and research.

4. Freemium Model

Individual users may access limited monthly usage for free, with optional premium upgrades for advanced features.

6.4 Cost Estimation

The main operational costs include:

- GPU cloud hosting for LLM inference
- API servers and load balancing
- Data storage and monitoring systems
- System maintenance and updates

GPU infrastructure represents the highest recurring cost. However, the use of parameter efficient fine-tuning techniques (such as LoRA) reduces retraining and computational expenses compared to full model training. Operational sustainability can be achieved if several mid-scale media organizations subscribe to the Professional Plan. For example, 10 organizations subscribing at mid-tier pricing can cover infrastructure and maintenance costs.

6.5 Pricing Strategy

Pricing must balance affordability within the Sri Lankan market and long-term sustainability.

A possible structure may include:

- Basic Plan – LKR 8,000 per month
- Professional Plan – LKR 25,000 per month
- Enterprise Plan – Custom pricing

Educational institutions may receive discounted pricing. API usage may follow per-request or per-token billing to ensure fairness based on consumption. The pricing strategy reflects the cost savings compared to hiring additional editorial staff while maintaining competitive affordability.

6.6 Competitive Advantage

SinhalaJournal-LLM offers:

- Dedicated support for Sinhala language
- Journalism-focused optimization
- Hybrid rule-based and LLM-based architecture
- Integration of multiple writing tools in one platform

Most existing solutions either lack Sinhala support or provide limited spell-checking features. This system fills a significant technological gap in the Sri Lankan media industry.

6.7 Intellectual Property Considerations

The intellectual property of the system includes:

- Fine-tuned Sinhala LLM model weights
- Hybrid grammar correction and style adaptation algorithms
- Journalism-specific datasets
- API integration and orchestration framework

The base large language model remains subject to its original licensing terms. Fine-tuned weights, integration logic, and custom-developed modules may be protected under institutional intellectual property policies. All deployments will comply with open-source licensing requirements and ethical AI usage standards.

7. BUDGET AND JUSTIFICATION

The estimated budget for the SinhalaJournal-LLM project reflects realistic infrastructure requirements for model fine-tuning, experimentation, multi-component deployment, and evaluation.

Since the system integrates multiple NLP components powered by a fine-tuned Sinhala Large Language Model (SinLlama) using Parameter-Efficient Fine-Tuning (LoRA), GPU-based computation is required during experimentation and training phases.

The budget also considers API hosting, system integration, testing, and demonstration deployment to represent the true economic scale of a production-ready AI system.

7.1 Hardware Costs

Table 2 Hardware Costs

Item	Estimated Cost (LKR)	Justification
Cloud GPU Rental (NVIDIA T4 / A100 equivalent)	120,000	Approx. 20 USD/hour × 40 hours fine-tuning & experimentation
Development Laptop	0	Existing personal device

Estimated Hardware Cost: 100,000 – 120,000 LKR

Justification:

Fine-tuning transformer-based models for multiple tasks (grammar correction, headline generation, style rewriting, summarization) requires GPU acceleration. Instead of purchasing a dedicated GPU (which would exceed 300,000 LKR), cloud GPU rental provides a cost-effective and scalable solution. Using Parameter-Efficient Fine-Tuning (LoRA) significantly reduces memory requirements and training time compared to full model retraining.

7.2 Software Licenses

Table 3 Software Licenses

Software	Cost (LKR)	Justification
Python	0	Open-source
PyTorch	0	Open-source
HuggingFace Transformers	0	Open-source

PEFT (LoRA)	0	Open-source
Docker	0	Open-source
VS Code	0	Open-source

Estimated Software Cost: 0 LKR

Justification:

The system relies entirely on open-source technologies, eliminating licensing costs.

7.3 Cloud and Hosting Services

Assumptions:

- vCPU, 4GB RAM instance
- 10,000 LKR per month
- months usage

Table 4 Cloud and hosting cost

Service	Monthly Cost (LKR)	Duration	Total Cost (LKR)
Cloud VM Hosting	10,000	6 months	60,000
Domain Registration	5,000	One-time	5,000

Estimated Cloud Cost: 65,000 LKR

Justification:

The cloud infrastructure hosts:

- REST API gateway
- Grammar validation module
- Headline generation module
- Style rewriting module
- Summarization module
- Demo web application
- Monitoring and logging services

Full LLM training does not run continuously. GPU usage is limited to the fine-tuning phase, while inference services operate on optimized deployments.

7.4 Human Effort

Although this is an academic project, the development effort can be estimated to demonstrate the project scale and economic value.

Table 5 Human Effort

Activity	Estimated Hours
Dataset preparation & annotation	80 hrs
Model fine-tuning & experimentation	120 hrs
System development & API integration	100 hrs
Evaluation & testing	60 hrs
Total Estimated Effort	360 hours

Cost Estimation

Assumptions:

- Intern rate = 1,500 LKR per day
- 1 working day = 8 hours

Total working days:

$$360 \text{ hours} \div 8 \text{ hours/day} = 45 \text{ days}$$

Estimated Human Effort Cost:

$$45 \text{ days} \times 1,500 \text{ LKR/day} = 67,500 \text{ LKR}$$

7.5 Total Estimated Direct Financial Cost

Table 6 Total Cost Estimation

Category	Estimated Cost (LKR)
Hardware (GPU Rental – if required)	0 – 32,000
Software	0
Cloud Hosting & Deployment	38,400 – 43,400
Human Effort (Intern basis)	67,500
Total Estimated Direct Cost	105,900 – 142,900 LKR

7.6 Overall Budget Justification

The proposed budget is intentionally minimized through:

- Use of institutional GPU infrastructure
- Adoption of open-source frameworks
- Parameter-Efficient Fine-Tuning (LoRA) to reduce computational overhead
- Limited-duration cloud hosting for deployment and demonstration

The primary financial cost arises from cloud hosting and development effort rather than expensive hardware acquisition.

This approach ensures financial feasibility while maintaining research quality, scalability potential, and technical robustness. The cost structure also demonstrates that SinhalaJournal-LLM is economically viable for both academic research and future commercialization.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Overview of Work Breakdown

The project is divided into the following major phases:

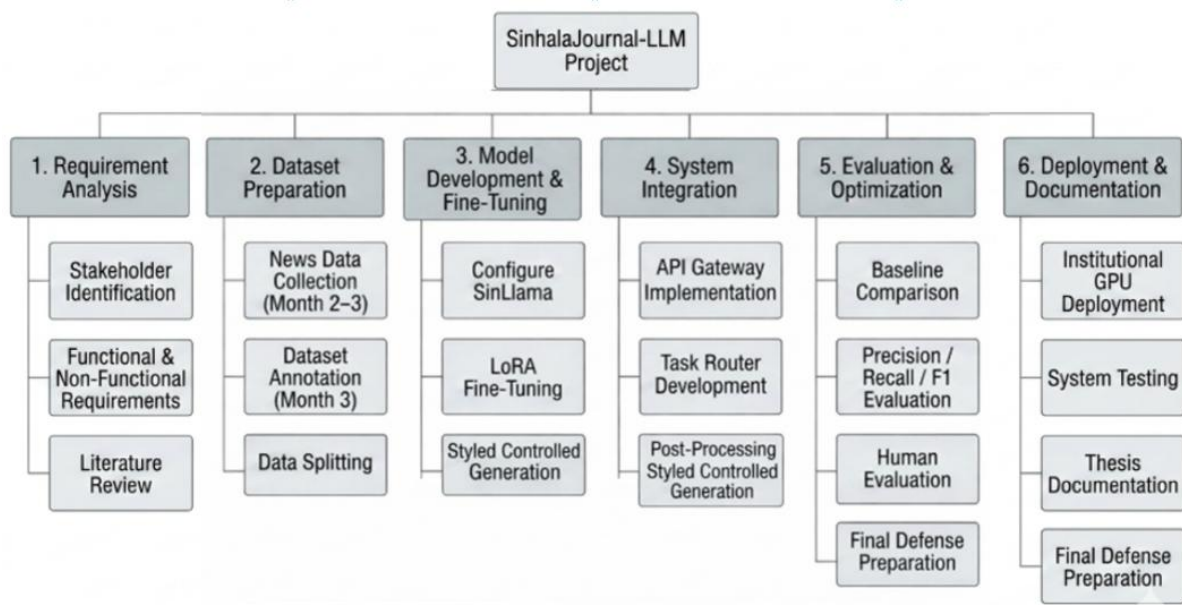


Figure 2 Work Breakdown Structure

Figure 2 presents the hierarchical Work Breakdown Structure (WBS) of the SinhalaJournal-LLM project. The project is organized into six major phases with clearly defined tasks to ensure structured development across the one-year timeline. The Sinhala Style rewriter dataset correction, LoRA based fine tuning, style controlled text generation and evaluation of semantic similarity and style accuracy—are highlighted to reflect the individual research contribution. The WBS also balanced workload distribution and coordinated system-level integration among the different NLP components of the SinhalaJournal-LLM.

8.2 Timeline Alignment

The project spans twelve months, following a phased and progressive development schedule:

- Months 1–2: Requirement analysis and literature review
- Months 2–3: Dataset collection
- Month 3: Dataset annotation
- Months 4–6: SinLlama fine-tuning and rule-based module development
- Months 6–7: Hybrid integration and system integration
- Months 7–9: Experimental evaluation and optimization
- Months 9–10: Institutional deployment and system testing
- Months 7–12: Documentation and thesis writing
- Months 10–12: Final validation and defense preparation

This timeline allows iterative experimentation, refinement, and integration while maintaining feasibility within the academic year.

8.3 Task Breakdown Per Member

Table 7 Task Breakdown per member

Phase	Member 1 – Grammar Checker	Member 2 – Headline Generator	Member 3 – Style Rewriter	Member 4 – News Summarizer
Requirement Analysis	Define objectives and evaluation metrics	Define generation objectives and style criteria	Define transformation objectives and tone categories	Define summarization objectives and evaluation criteria
Dataset Preparation	Prepare annotated error–correction dataset	Prepare labeled headline dataset	Prepare style transformation dataset	Prepare summarization dataset
Model Development	Implement LoRA fine-tuning and rule-based validation	Implement style-conditioned fine-tuning	Implement instruction-based fine-tuning	Implement summarization model
System Integration	Develop API and integrate with router	Develop API and integrate with router	Develop API and integrate with router	Develop API and integrate with router
Evaluation	Evaluate using Precision, Recall, F1, GLEU	Evaluate relevance and style consistency	Evaluate tone preservation and consistency	Evaluate using ROUGE and baseline comparison

8.4 Balanced Workload Distribution

The Workload distribution of the SinhalaJournal-LLM project is structured to ensure fairness and balanced technical contribution among all team members. Each member is responsible for developing their component within the shared platform.

The Sinhala Style Rewriter Component , Which is the focus of this , involves several key technical tasks including data preparation , Style annotation , LoRA based fine tuning , and evaluation using matrices. This component need careful dataset construction to ensure that rewritten articles maintain semantic meaning while adapting to the selected journalistic style.

9. GANTT CHART

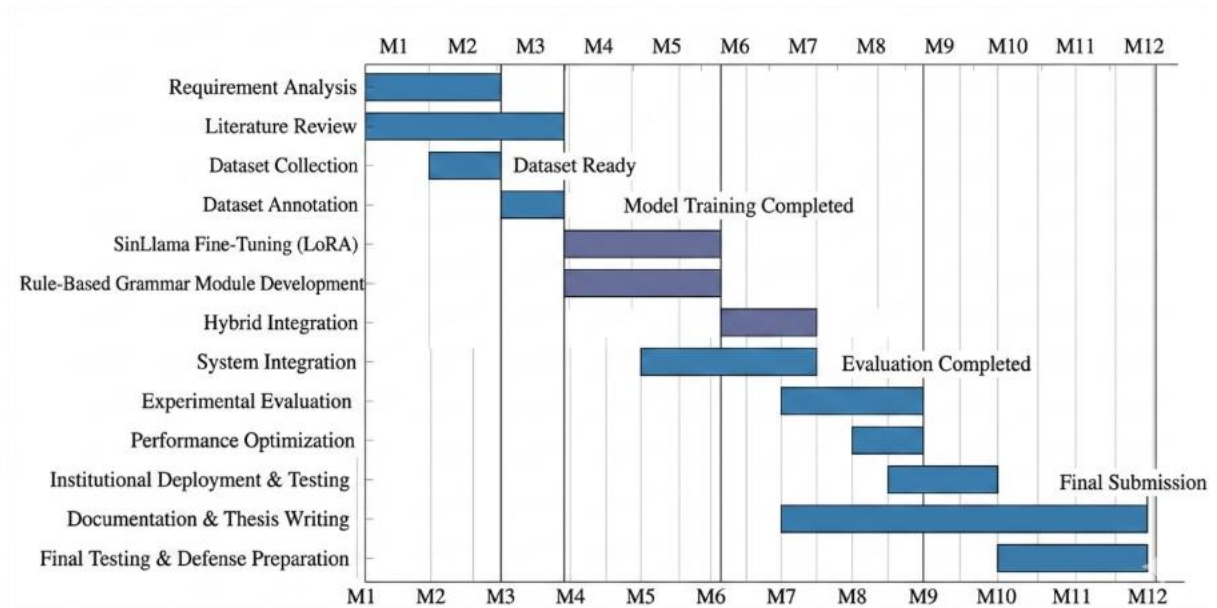


Figure 3 12 month project timeline Gantt Chart

[Figure 3](#) presents the 12-month project timeline in the form of a Gantt chart. The schedule aligns research, development, integration, and evaluation phases in a structured and progressive manner. Early months focus on requirement analysis and dataset preparation, followed by model fine-tuning and rule-based module development. System integration and experimental evaluation occur in the middle phase, while deployment, documentation, and final validation are scheduled in the concluding months.

Milestones are clearly defined to monitor progress, including dataset completion, model training completion, evaluation finalization, and final submission. The timeline ensures feasibility, iterative refinement, and sufficient preparation for project defense.

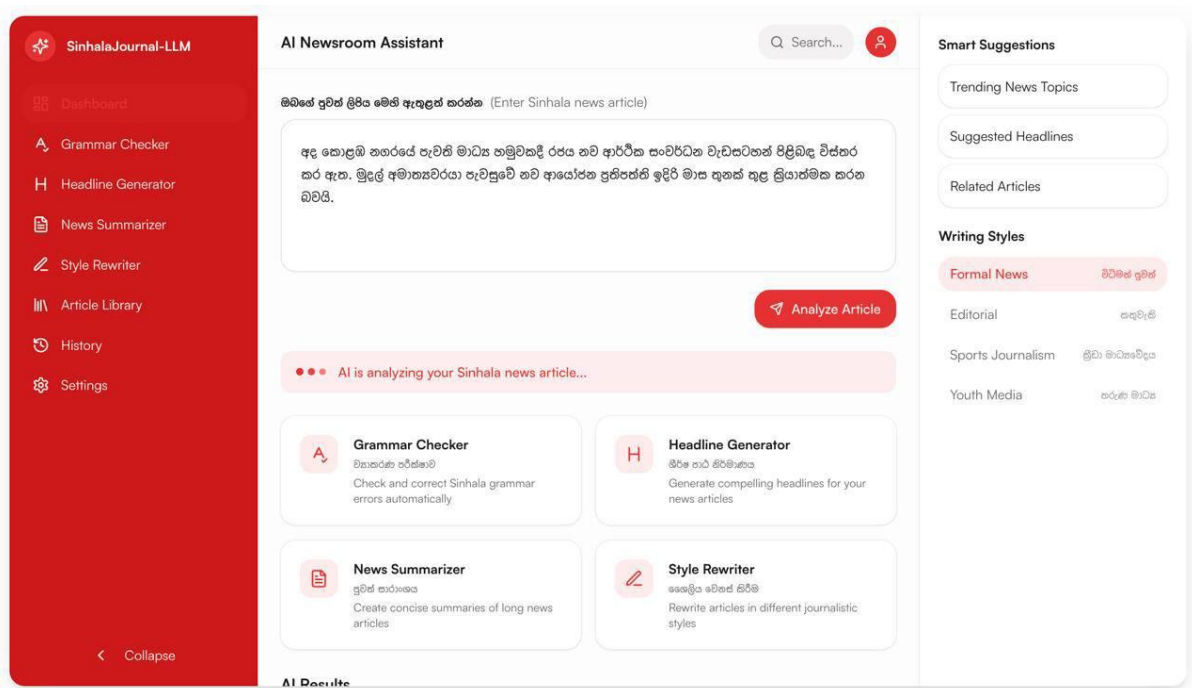
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11.APPENDICES

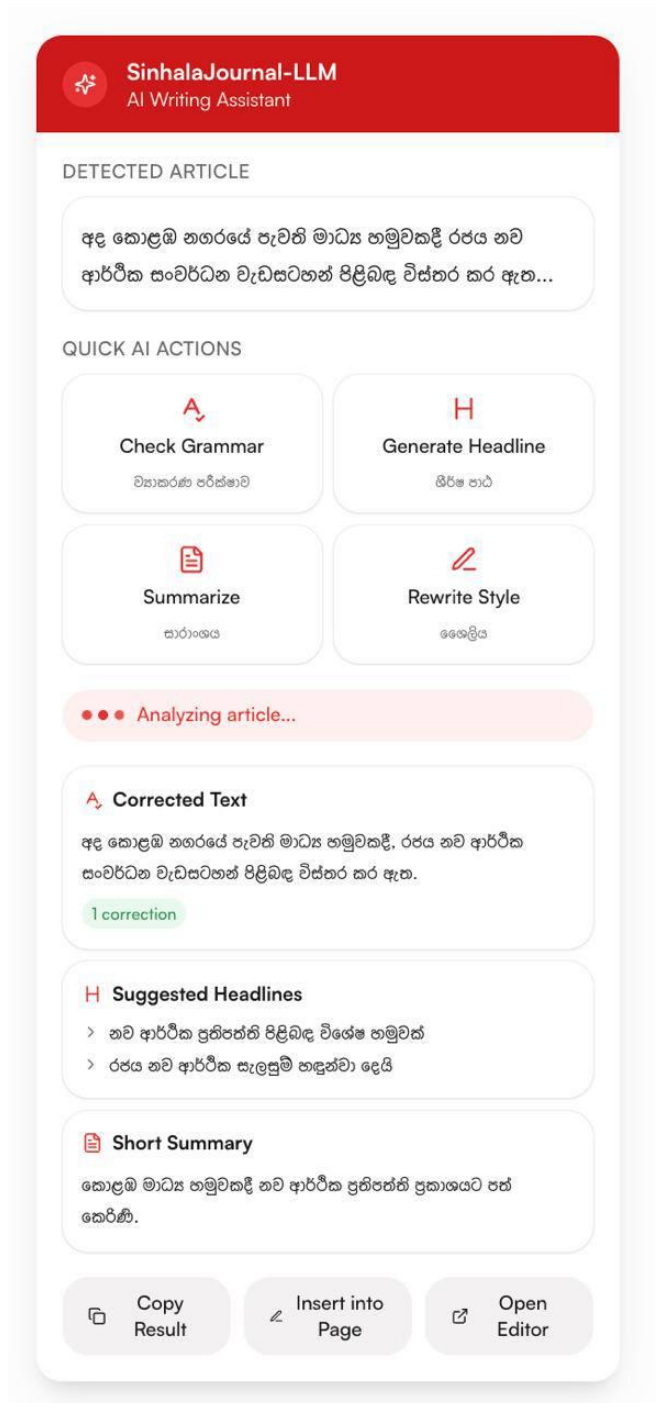
11.1 Preliminary UI mockups

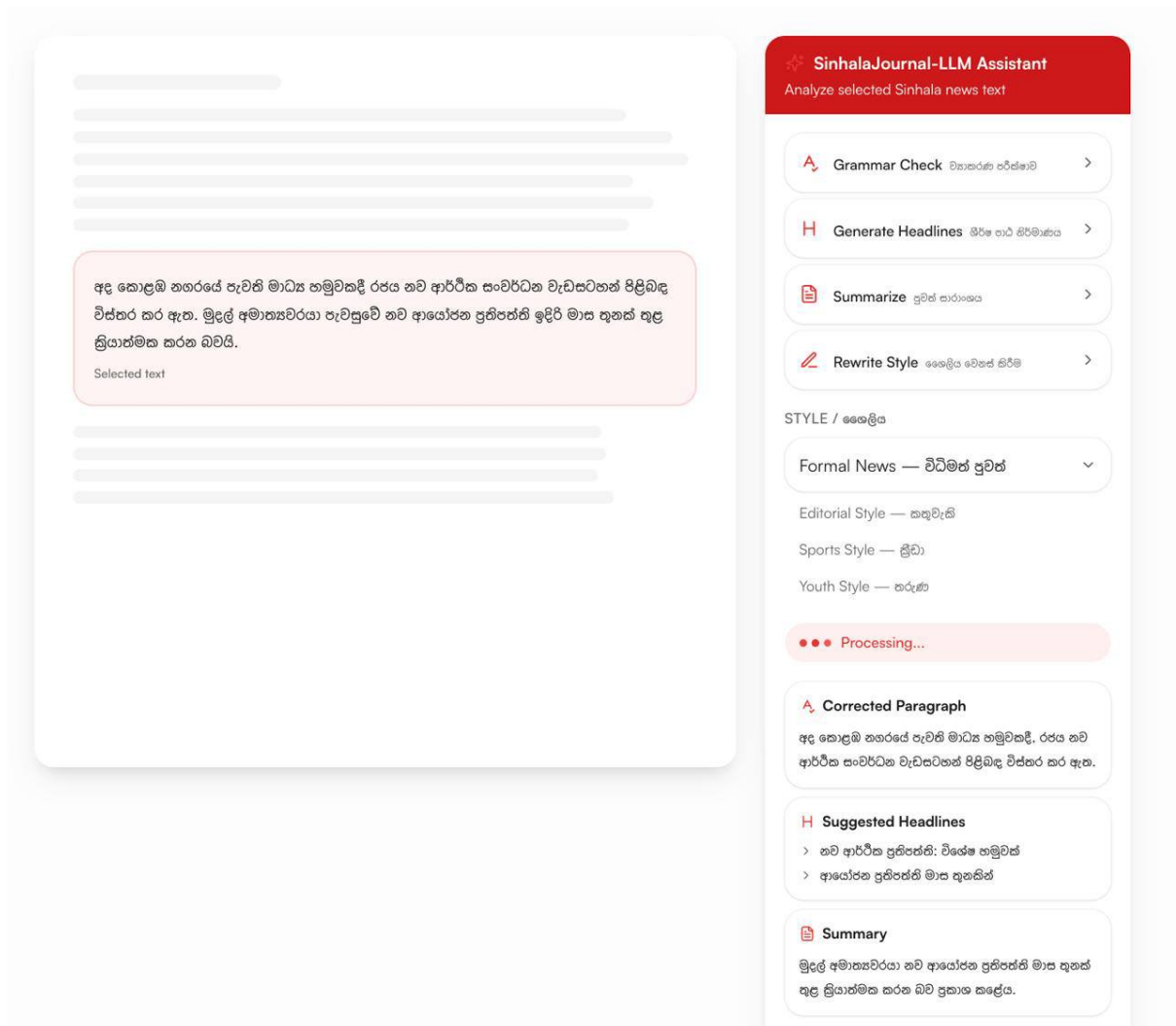
Web UI



Appendix 1 Preliminary UI mockups

Web Extension UI





11.2 Risk analysis

The development of the Sinhala Style Rewriter component in the SinhalaJournal-LLM system has several risks that may affect to performance, data quality and feasibility.

1. Technical Risk

The style rewriter produce outputs that alter the original mean of the news article or generate grammatically incorrect Sinhala text.

Mitigation - Semantic similarity evaluation , grammar validation and human review will be used to ensure that rewritten outputs keep the original mean.

2. Style Consistency Risk

The generated news article output not fully follow the Selected Style. (Formal , editorial , youth oriented)

Mitigation – Use Style-specific prompts and training datasets during fine tuning.

3. Data Quality Risk

Collected Sinhala news article will contain noisy data, duplicate details, or inconsistent formatting due to automated web scraping.

Mitigation – Data preprocessing steps like Unicode normalization , duplicate removal , text cleaning will be apply to increase dataset quality.

4. Model Bias Risk

Training data collected from specific news sources may contain political , social biases that could influence the model’s generated outputs.

Mitigation – The dataset will be collected form multiple news websites with different topics.

5. Risk to Infrastructure

During the research process, limited GPU resources may make it difficult to conduct large-scale experiments or complete model training.

Mitigation: To lower computational requirements and enable effective model adaptation with constrained resources, parameter-efficient fine-tuning techniques like Low-Rank Adaptation (LoRA) will be employed.